SYMMETRIES OF THE COLLISION DIGIT FUNCTION

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Abstract. We prove three structural theorems for the digit function $d_n(a) = \lfloor (n+1)a/m \rfloor - \lfloor na/m \rfloor$ at base 3. First, a *slice palindrome*: $d_{m-1-n}(a) = d_n(a)$ for every coprime a and every interior slice. Second, a *full-range constant*: the left-right crossing imbalance across all m slices is exactly 1 for every coprime class. Third, a *middle block balance*: the central third of the slice system carries zero left-right imbalance, forced by the palindrome. At $\ell = 2$, we further prove that the crossing imbalance $f_j(a) = C_j(2, a) - C_j(0, a)$ is invariant under multiplicative inversion ($f_j(a) = f_j(a^{-1})$), and verify this through 5 lags. Since $\chi(a^{-1}) = \overline{\chi(a)}$, the inversion invariance (if it extends to all lags) would force $\sum \chi(a)\, f_j(a) \in \mathbb{R}$ for every odd Dirichlet character, yielding the covariance identity $K_\ell(0, 1) = \frac{1}{4} K_\ell(1, 1)$ via Parseval. At general bases, the collision mean value theorem $V^2_{\text{rms}} = O(b^\ell)$ reduces to a per-bin covariance bound (Conjecture 6.12), verified at bases $3, 7, 10$ through 6 lags. No L-function theory enters the argument.

1. Introduction

The collision periodic table $\mathcal{T}_\ell(a)$ records the signed collision count at lag ℓ for primes $p \equiv a \pmod{b^{\ell+1}}$. By finite determination [1], $\mathcal{T}_\ell$ is a well-defined function on the coprime classes modulo $m = b^{\ell+1}$.

At lag 1 ($m = b^2$): the table has $\varphi(b^2)$ entries, and the collision PNT [4] gives $\sum S^\circ(p) = O(x \exp(-c\sqrt{\log x}))$. At higher lags: the table refines, the modulus grows, and the collision counts probe deeper digit correlations.

This paper studies the *true spectral ladder*: lag ℓ at modulus $b^{\ell+1}$. The centered collision variance $V^2_{\text{rms}} = (1/\varphi) \sum |S^\circ_\ell(a)|^2$ grows with ℓ. The question: how fast?

The expected growth is $V^2_{\text{rms}}(\ell) = O(b^\ell)$. At general bases, the mean value theorem reduces to a per-bin covariance bound (Conjecture 6.12), verified for bases $3, 7, 10$ through the tested range.

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At base 3, the story is sharper. The digit function $d_n(a) = \lfloor (n+1)a/m \rfloor - \lfloor na/m \rfloor$ turns out to be palindromic ($d_{m-1-n} = d_n$ for interior n), and the crossing imbalance between left and right bin positions is invariant under multiplicative inversion of the class ($f_j(a) = f_j(a^{-1})$). Assuming the inversion invariance extends to all lags (proved at the base case, verified through 5 lags), a Dirichlet character decomposition and Parseval's identity give the covariance identity $K(0,1) = K(1,1)/4$, which would reduce the base-3 problem to a single second-moment estimate. No L-function theory enters the proof architecture.

2. The True Ladder

Definition 2.1. The *true spectral ladder* at base b is the sequence of periodic tables $\mathcal{T}_1, \mathcal{T}_2, \mathcal{T}_3, \ldots$ where $\mathcal{T}_\ell$ is defined on $(\mathbb{Z}/b^{\ell+1}\mathbb{Z})^*$ using the lag-ℓ collision count $C(b^\ell \bmod p) - \lfloor (p-1)/b \rfloor$.

Observation 2.2 (The wrong ladder vs. the right ladder)**.** The naive ladder (fixed lag 1, increasing modulus $b^2, b^3, b^4, \ldots$) is self-similar: the potential repeats the $\varphi(b^2)$ values at every rung. The true ladder (lag ℓ at $m = b^{\ell+1}$) has genuinely new structure:

ℓ	m	φ	V_{rms}	$V_{\max}$	distinct
1	100	40	2.58	7.3	20
2	1000	400	9.87	88.0	78
3	10000	4000	32.1	897.8	268
4	100000	40000	101.9	9000	910

$V_{\text{rms}}/b^{\ell/2}$ converges: $0.82, 0.99, 1.01, 1.02$. $V_{\max}/b^\ell$ converges to $(b-1)/b = 0.9$.

3. The Extreme Class

Observation 3.1 (Repunit extreme)**.** Computation at bases $3, 7, 10, 12$ through 10 lags (22 cases) gives exactly $\mathcal{T}_\ell(b^\ell - 1) = (b-1)b^{\ell-1} - 1$, with complement $\mathcal{T}(b^{\ell+1} - b^\ell + 1) = -(b-1)b^{\ell-1}$.

Remark 3.2. The MVT proof (Theorem 7.2) uses only $|\mathcal{T}(b^\ell - 1)| = O(b^\ell)$, which follows from the range bound $|\mathcal{T}| \le b^\ell - 1$ of [1]. The exact formula sharpens the constant but is not required for the order of growth.

Corollary 3.3. $V_{\max} \le b^\ell - 1$ *(the range bound of* [1]*). Computation gives* $V_{\max} = (b-1)/b \cdot b^\ell + O(1)$.

4. The Recursion

Proposition 4.1 (Lag recursion)**.** *$\mathcal{T}_\ell(a) = \mathcal{T}_{\ell-1}(a \bmod b^\ell) + \delta_\ell(a)$, where δ_ℓ is the lag-ℓ correction.*

Proof. By finite determination, $\mathcal{T}_\ell(a)$ depends on $a \bmod b^{\ell+1}$ and $\mathcal{T}_{\ell-1}(a \bmod b^\ell)$ depends on $a \bmod b^\ell$. The difference $\delta_\ell(a)$ captures the information in the finer residue class. □

The correction δ_ℓ measures the net change in collisions when the multiplier increases from $b^{\ell-1}$ to b^ℓ. Writing $s = b^{\ell-1}r \bmod p$: the lag-$(\ell-1)$ collision compares $\mathrm{digit}(r)$ to $\mathrm{digit}(s)$, while the lag-ℓ collision compares $\mathrm{digit}(r)$ to $\mathrm{digit}(bs \bmod p)$.

Definition 4.2. The *change set* is $Z = \{s : \mathrm{digit}(s) \neq \mathrm{digit}(bs \bmod p)\}$. For $s \in Z$ with $\mathrm{digit}(s) = d$, $\mathrm{digit}(bs) = d'$, and preimage $r = s \cdot (b^{\ell-1})^{-1} \bmod p$ with $\mathrm{digit}(r) = d_r$:

- $d_r = d$: a disrupted collision. Contribution -1 to δ.
- $d_r = d'$: a created collision. Contribution $+1$ to δ.
- $d_r \notin \{d, d'\}$: no change. Contribution 0.

5. The Per-Bin Flow Bound

Definition 5.1. For each bin $d \in \{0, \ldots, b-1\}$, the *bin flow* is $F(d) = \mathrm{out_match}(d) - \mathrm{in_match}(d)$, where $\mathrm{out_match}(d)$ counts s values entering bin d (under $s \to bs$) with $\mathrm{digit}(r) = d$, and $\mathrm{in_match}(d)$ counts s leaving bin d with $\mathrm{digit}(r) = d$.

Definition 5.2. For bins $d \neq d'$, define the *per-source-bin flow*:

$$\begin{aligned} F(d, d') = {} & \#\{r \in B_d : \mathrm{digit}(s) = d',\ \mathrm{digit}(bs) = d\} \\ & - \#\{r \in B_d : \mathrm{digit}(s) = d,\ \mathrm{digit}(bs) = d'\} \end{aligned}$$

with $s = b^{\ell-1}r \bmod p$, and the *per-bin flow* $F(d) = \sum_{d' \neq d} F(d, d')$. Then $\delta_\ell(a) = \sum_d F(d)$.

Lemma 5.3 (Per-source-bin bound at $\ell = 2$)**.** *At lag 2: $|F(d, d')| \leq 1$ for every base $b \geq 3$, every coprime class a, and every pair $d \neq d'$. Consequently $|F(d)| \leq b - 1$ and $|\delta| \leq b(b-1)$.*

Proof. At $\ell = 2$: for $r \in B_d$, $\lfloor br/p \rfloor = d$, so $br \bmod p = br - dp \in [0, p)$. The images $\{br - dp : r \in B_d\}$ form a contiguous arithmetic progression with step b in $[0, p)$. The count of a step-b progression in any interval of width W is $\lfloor W/b \rfloor$ or $\lfloor W/b \rfloor + 1$ (elementary). The transition sets $T(d' \to d)$ and $T(d \to d')$ are contiguous arcs of width $\lfloor p/b^2 \rfloor$ or $\lfloor p/b^2 \rfloor + 1$. The counts of the step-b lattice in two such arcs differ by at most 1. □

Observation 5.4 (Lag 2)**.** Verified at base 10, all 400 classes and all 90 pairs (36000 checks): $\max|F(d,d')| = 1$. The maximum $|F(d)| = 9 = b-1$ is achieved only at the repunit ($F(d,d') = +1$ for all $d' \neq d$).

Observation 5.5 (Higher lags)**.** At $\ell \geq 3$: the lattice wraps and $\max|F(d,d')|$ grows as $b^{\ell-2}$. However, the *average* $\langle F(d,d')^2\rangle$ over all classes remains moderate:

ℓ	$\max\lvert F\rvert$	$\langle F^2\rangle$
2	1	0.32
3	b	3.83

The normalized correction energy $\langle\delta^2\rangle/\varphi^2 = (1/\varphi^2)\sum_a \delta_\ell(a)^2$ converges to a constant depending only on b:

ℓ	$\langle\delta^2\rangle/\varphi^2$ (base 10)	(base 3)
2	0.276	0.667
3	0.285	0.623
4	0.284	0.608
8	—	0.592

At base 10: the ratio stabilizes at 0.284 by lag 3. At base 3: it converges to 0.592 by lag 8. The extreme classes (repunit family) carry $O(b^\ell)$ per lag; the bulk carries $O(1)$. The convergence reflects the palindromic cycle structure: the $b-1$ radials ($\sigma(a)$ mod $(b-1)$, preserved by multiplication by b) organize the corrections into a repeating pattern whose average energy per class is bounded.

Observation 5.6. Verified at base 10, lag 2, all 400 classes and 10 bins (4000 checks): $\max|F(d)| = 9 = b-1$, achieved only at the repunit class $a = 99$. Distribution: $|F| = 0$ (43.9%), $|F| = 1$ (39.1%), $|F| \leq 2$ (92.4%).

6. The Bulk Bound

Proposition 6.1 (Deterministic bulk bound at $\ell = 2$)**.** *At lag 2: $|\delta_2(a)| = |\sum_d F(d)| \leq \sum_d |F(d)| \leq b(b-1)$ for every class a, by Lemma 5.3.*

For the repunit class: $F(d,d') = +1$ for all $d' \neq d$, giving $|F(d)| = b-1$ for all d and $|\delta| = b(b-1)$. Over ℓ lags: δ accumulates to $(b-1)b^{\ell-1}$ (Observation 3.1).

For generic classes: the $F(d,d')$ have mixed signs, giving $|\delta| \ll 2b(b-1)$. Computation shows 92.5% of classes at lag 2 have $|\delta| \leq b$.

Definition 6.2. If the limit exists, define the *collision constant*

$$c_\delta(b) = \lim_{\ell\to\infty} \frac{1}{\varphi(b^{\ell+1})^2} \sum_{\substack{a \bmod b^{\ell+1} \\ \gcd(a,b)=1}} \delta_\ell(a)^2.$$

Computation suggests $c_\delta(10) \approx 0.284$ and $c_\delta(3) \approx 0.58$.

The vortex decomposition. The per-source-bin flow $F(d, d')$ decomposes by *gap* $g = (d - d') \bmod (b-1)$. Since $b \equiv 1 \pmod{b-1}$, multiplication by b preserves the digit-sum residue $\sigma(r) \bmod (b-1)$, which identifies the bin index d with radial index $d \bmod (b-1)$. The flow splits into two components:

Radial flow (gap 0): flow between bins $d \equiv d' \pmod{b-1}$, i.e., between bins 0 and $b-1$ (which share radial 0). This channel carries the repunit accumulation.

Transverse flow (gaps $1, \ldots, b-2$): flow across distinct radials. This is the dominant component.

Observation 6.3 (Vortex structure)**.** At base 10, lag 3: the transverse flow carries 97.6% of all nonzero $F(d, d')$ entries; the radial flow (gap 0) carries 2.4%. The average $|F|$ is approximately equal across all nine gap values (1.76–1.83). The transverse flow has *mixed signs* across classes, producing cancellation in δ; the radial flow has *coherent sign* at the extreme classes, producing accumulation.

The polarity alternation. The cancellation in the transverse flow is forced by the *polarity structure* of the digit function. The b bins partition into three zones of alternating polarity (visible in the numeric polarity chart of [7]):

- *Positive polarity*: bins where collisions are preferentially created.
- *Negative polarity*: bins where collisions are preferentially destroyed.
- *Neutral*: the zone containing bin $b-1$ (the repunit bin), where creation and destruction coexist.

The zones alternate around the base: each positive bin is adjacent to a negative bin. For generic classes, each image transition (s changing bin under $\times b$) generates a $+1$ creation in one entry of $F(d, d')$ and a -1 disruption in another. The alternating polarity ensures these contributions have mixed signs across bin pairs, preventing coherent accumulation.

Lemma 6.4 (Flow conservation)**.** *For each bin d: the number of images entering bin d under $s \mapsto bs \bmod p$ equals the number leaving. That is, $N_{\text{in}}(d) = N_{\text{out}}(d)$.*

Proof. The map $s \mapsto bs \bmod p$ is a permutation of $\{1, \ldots, p-1\}$. The bin occupancy $\#\{s : \lfloor bs/p \rfloor = d\}$ is therefore the same before and after the permutation. Each element that changes bin contributes one departure and one arrival; the totals match at each bin. □

Lemma 6.5 (Per-arc source uniformity)**.** *Let $g = b^{\ell-1} \bmod p$ and $h = g^{-1} \bmod p$. The change set $Z = \{s : \mathrm{digit}(s) \neq \mathrm{digit}(bs)\}$ consists of $b-1$ arcs of width $W \approx 2p/b^2$ near the bin boundaries. The preimage of each arc under $r = sh \bmod p$ is an arithmetic progression of common difference h and W terms.*

By the three-gap theorem: the count of elements of a single arithmetic progression (step h, W terms) in any interval of width p/b is $W/b+O(1)$.

Proof. An arithmetic progression of W terms with step h on a circle of circumference p places at most $\lfloor W/\lfloor p/h \rfloor \rfloor + 1$ terms in any interval of width p/b, which is $W/b + O(1)$ when $\gcd(h, p) = 1$. □

Per-bin overlap covariance.

Lemma 6.6 (Finite determination for per-bin overlaps)**.** *For fixed b, ℓ, d, the restricted collision count $C_{\ell,d}(p) = \#\{r \in B_d : \mathrm{digit}(b^\ell r \bmod p) = d\}$ depends only on $p \bmod b^{\ell+1}$ for all sufficiently large primes p. This follows from the finite-determination theorem of* [1]*, since $C_{\ell,d}$ is a sub-sum of the collision count $C(b^\ell \bmod p)$ restricted to a single source bin.*

Definition 6.7 (Per-bin overlap shift)**.** For each coprime class a mod $b^{\ell+1}$ and bin $d \in \{0, \ldots, b-1\}$, define

$$\Delta_{a,d} = C_{\ell,d}(p) - C_{\ell-1,d}(p),$$

where p is any sufficiently large prime with $p \equiv a \pmod{b^{\ell+1}}$. By Lemma 6.6, this is well-defined as a function of the class a.

Lemma 6.8 (Per-bin decomposition)**.** $\delta_\ell(a) = \sum_{d=0}^{b-1} \Delta_{a,d}$.

Proof. Regroup the created and disrupted collisions by source bin. □

Proposition 6.9 (Bin-level complement symmetry)**.** *For every coprime class a mod $b^{\ell+1}$:*

$$\Delta_{a,d} = \Delta_{a,\,b-1-d} \qquad (0 \le d \le b-1).$$

Proof. Let p be any representative prime for class a. The complement map $r \mapsto p - r$ is a bijection of $\{1, \ldots, p-1\}$ that sends $B_d \to B_{b-1-d}$ (since $\mathrm{digit}(p-r) = b-1-\mathrm{digit}(r)$) and satisfies $\mathrm{digit}(g(p-r) \bmod p) =$

$b-1-\operatorname{digit}(gr \bmod p)$ for any multiplier g. Therefore

$$\begin{aligned} M_d^g &:= \#\{r \in B_d : \operatorname{digit}(gr) = d\} \\ &= \#\{r \in B_{b-1-d} : \operatorname{digit}(gr) = b-1-d\} = M_{b-1-d}^g. \end{aligned}$$

Since this holds for both $g = b^{\ell-1}$ and $g = b^\ell$: $\Delta_{a,d} = M_d^{b^\ell} - M_d^{b^{\ell-1}} = M_{b-1-d}^{b^\ell} - M_{b-1-d}^{b^{\ell-1}} = \Delta_{a,b-1-d}$. By Lemma 6.6, the identity holds at the class level. □

Corollary 6.10 (Bilateral covariance symmetry)**.** *The covariance matrix $K_\ell(d,d') = \sum_a \Delta_{a,d}\,\Delta_{a,d'}$ satisfies $K_\ell(d,d') = K_\ell(b-1-d,\, b-1-d')$.*

Proof. By Proposition 6.9: $\Delta_{a,d}\,\Delta_{a,d'} = \Delta_{a,b-1-d}\,\Delta_{a,b-1-d'}$. Sum over a. □

At base 3: Proposition 6.9 gives $\Delta_{a,0} = \Delta_{a,2}$ for every class, so the covariance matrix has only two free parameters: $K(0,0)$ and $K(0,1)$.

Lemma 6.11 (Quadratic majorization)**.** *For every class a: $\delta_\ell(a)^2 = (\sum_d \Delta_{a,d})^2 \le b \sum_d \Delta_{a,d}^2$. Hence $\sum_a \delta_\ell(a)^2 \le b \sum_a \sum_d \Delta_{a,d}^2$.*

Proof. Cauchy–Schwarz applied to the vector $(\Delta_{a,0}, \ldots, \Delta_{a,b-1})$. □

Conjecture 6.12 (Per-bin covariance bound)**.** For each base $b \ge 3$: $\sum_{d,d'} K_\ell(d,d') = O(b^{2\ell})$, where $K_\ell(d,d') = \sum_a \Delta_{a,d}\,\Delta_{a,d'}$.

Conjecture 6.12 is verified at bases $3, 7, 10$ through 6 lags. At base 3, the mechanism is identified in Theorem 6.31.

Lemma 6.13 (Base-3 overlap reduction)**.** *At base 3, for every class a there exist integers x_a, y_a such that $(\Delta_{a,0}, \Delta_{a,1}, \Delta_{a,2}) = (x_a, y_a, x_a)$. Hence $\delta_\ell(a) = 2x_a + y_a$ and $\sum_d \Delta_{a,d}^2 = 2x_a^2 + y_a^2$.*

Proof. Set $x_a = \Delta_{a,0}$, $y_a = \Delta_{a,1}$. By Proposition 6.9: $\Delta_{a,2} = \Delta_{a,0} = x_a$. □

Base-3 covariance identity.

Definition 6.14 (Boundary decomposition)**.** Write the middle-bin indicator as $1_I = 1_A - 1_B$ where $1_A(r) = 1_{\operatorname{digit}(r)\ge 1}$ and $1_B(r) = 1_{\operatorname{digit}(r)\ge 2}$. For a multiplier g, define

$$\begin{aligned} L_g(a) &= \sum_r 1_A(r)\,1_I(gr) - \sum_r 1_B(r)\,1_I(gr), \\ R_g(a) &= \sum_r 1_I(r)\,1_A(gr) - \sum_r 1_I(r)\,1_B(gr). \end{aligned}$$

Set $L_a = L_{3^\ell}(a) - L_{3^{\ell-1}}(a)$ and $R_a = R_{3^\ell}(a) - R_{3^{\ell-1}}(a)$.

Lemma 6.15 (Boundary decomposition identity)**.** *$y_a = L_a + R_a$ for every class a.*

Proof. By the indicator expansion, $Y_g(a) = L_g(a) + R_g(a)$ for each multiplier g. Taking the lag difference gives $y_a = L_a + R_a$. □

Definition 6.16. For a multiplier g, the *cross-bin count* is $T_g(a) = \#\{r : \mathrm{digit}(r) = 2,\ \mathrm{digit}(gr) = 1\}$.

Theorem 6.17 (Entrywise transition-matrix antisymmetry)**.** *For every base $b \geq 3$, lag $\ell \geq 2$, bins d, d', and coprime class $a \bmod b^{\ell+1}$:*

$$\Delta M_{d,d'}(a) = -\Delta M_{d,d'}(m-a),$$

where $\Delta M_{d,d'} = M_{d,d'}(b^\ell) - M_{d,d'}(b^{\ell-1})$ and $M_{d,d'}(g) = T_{g,d\to d'}(a)$. In particular, $\Delta_{a,d} = -\Delta_{m-a,d}$ for the diagonal entries.

Proof. By the finite-determination theorem of [1]: on the top-level slice system $n \in \{0, \ldots, m-1\}$ with $m = b^{\ell+1}$, the source digit on slice n is $\lfloor n/b^\ell \rfloor$ and the target digit at multiplier b^ℓ is $n \bmod b$. The target digit at multiplier $b^{\ell-1}$ is $\lfloor (n \bmod b^2)/b \rfloor$.

Therefore $M_{d,d'}(b^\ell)$ sums over the slice subset $S^{(\ell)}_{d\to d'} = \{n : \lfloor n/b^\ell \rfloor = d,\ n \bmod b = d'\}$, and $M_{d,d'}(b^{\ell-1})$ sums over $S^{(\ell-1)}_{d\to d'} = \{n : \lfloor n/b^\ell \rfloor = d,\ \lfloor (n \bmod b^2)/b \rfloor = d'\}$. Both have cardinality $b^{\ell-1}$.

By the reflection identity of [1]: $d_n(m-a) = 1-d_n(a)$ for every interior slice. Applying this to each subset with representatives $p = mt + a$ and $p' = mt + (m-a)$ sharing the same quotient t:

$$M_{d,d'}(b^k; a) + M_{d,d'}(b^k; m-a) = |S^{(k)}_{d\to d'}|(2t+1) + \text{endpoint}$$

for $k = \ell$ and $k = \ell-1$. Since $|S^{(\ell)}_{d\to d'}| = |S^{(\ell-1)}_{d\to d'}| = b^{\ell-1}$: the right side is the same for both lags. Therefore $\Delta M_{d,d'}(a) + \Delta M_{d,d'}(m-a) = 0$. □

Remark 6.18. This is the universal entrywise antisymmetry: the complement class sees the opposite transition increment at every bin pair, not just in total. The total antisymmetry $\delta(a) + \delta(m-a) = 0$ is the trace. The per-bin diagonal version $\Delta_{a,d} = -\Delta_{m-a,d}$ is a special case. The full entrywise version gives the conservation law and, combined with the character-level analysis of Theorem 6.30, the base-3 covariance identity.

Proposition 6.19 (Boundary oddness)**.** *At base 3: $L_a = -L_{m-a}$ and $R_a = -R_{m-a}$ for every class a.*

Proof. By Theorem 6.17: $\Delta_{a,d} = -\Delta_{m-a,d}$ for every bin d. Since $y_a = L_a + R_a$: $y_{m-a} = -y_a$, hence $L_{m-a} + R_{m-a} = -(L_a + R_a)$. Since $z_a = L_a - R_a$ (the boundary asymmetry): $z_{m-a} = -z_a$, hence

$L_{m-a} - R_{m-a} = -(L_a - R_a)$. Adding: $L_{m-a} = -L_a$. Subtracting: $R_{m-a} = -R_a$. □

Remark 6.20. Boundary oddness gives $\sum L_a^2 = \sum L_{m-a}^2$ and $\sum R_a^2 = \sum R_{m-a}^2$ (trivially, by relabeling). It does *not* imply $\sum L^2 = \sum R^2$ or the endpoint-energy equality $\sum u^2 = \sum w^2$. Those require the deeper character-level mechanism of Theorem 6.30.

Theorem 6.21 (Slice palindrome)**.** *For every coprime class a and every interior slice ($0 < n < m-1$): $d_{m-1-n}(a) = d_n(a)$.*

Proof. For a coprime to m, neither na/m nor $(n+1)a/m$ is an integer when $0 < n < m-1$. For non-integer x: $\lceil x \rceil = \lfloor x \rfloor + 1$. Therefore $d_{m-1-n}(a) = \lceil (n+1)a/m \rceil - \lceil na/m \rceil = \lfloor (n+1)a/m \rfloor - \lfloor na/m \rfloor = d_n(a)$. □

Theorem 6.22 (Full-range imbalance)**.** *For every coprime class a:*

$$f^{\text{full}}(a) := \#\{n \in [0,m) : d_0(n){=}2,\ d_n(a){=}1\} \\ - \#\{n \in [0,m) : d_0(n){=}0,\ d_n(a){=}1\} \ = \ 1.$$

Proof. The crossing set is $S(a) = \{n : na \bmod m \in [m{-}a,\, m)\}$. Substituting $c = na \bmod m$: $f^{\text{full}}(a) = \#\{c \in [m{-}a, m) : d_0(a^{-1}c \bmod m) = 2\} - \#\{c \in [m{-}a, m) : d_0(a^{-1}c \bmod m) = 0\}$. Since $d_0(a^{-1}c \bmod m) = (a^{-1} \bmod 3)(c \bmod 3) \bmod 3$, the map $c \bmod 3 \mapsto d_0$ is a fixed permutation π_a of $\{0,1,2\}$ that sends $0 \mapsto 0$ (for both $a \equiv 1$ and $a \equiv 2 \pmod 3$). In $[m{-}a,\, m)$ with $m \equiv 0 \pmod 3$, direct counting gives one more $c \equiv 2$ than $c \equiv 0$. Under π_a: for $a \equiv 1$, residue 2 maps to $d_0 = 2$; for $a \equiv 2$, residue 2 maps to $d_0 = 1$ and residue 1 (which also has one extra) maps to $d_0 = 2$. In both cases $\#\{d_0 = 2\} - \#\{d_0 = 0\} = 1$. □

Theorem 6.23 (Middle source block balance)**.** *The middle source block $B_1 = \{n : 3^\ell \le n < 2 \cdot 3^\ell\}$ satisfies $f^{B_1}(a) = 0$ for every coprime a.*

Proof. The digit complement $\tau(n) = m{-}1{-}n$ maps B_1 to itself (since B_1 is centered at $m/2$). Within B_1, all slices are interior, so Theorem 6.21 gives $d_{\tau(n)}(a) = d_n(a)$ for every $n \in B_1$. Since τ swaps $d_0 = 0 \leftrightarrow d_0 = 2$: $f^{B_1}(a) = \sum_{n \in B_1} \varepsilon(n)\, d_n(a) = \sum_{n \in B_1} \varepsilon(\tau(n))\, d_{\tau(n)}(a) = -\sum_{n \in B_1} \varepsilon(n)\, d_n(a) = -f^{B_1}(a)$, hence $f^{B_1}(a) = 0$. □

Definition 6.24 (Character spectrum and mismatch field)**.** Let $\tau\colon n \mapsto 3^\ell{-}1{-}n$ be the digit complement on slices. For each odd Dirichlet character χ modulo m, define the *character spectrum* $D_n(\chi) = \sum_{a \text{ coprime}} \chi(a)\, d_n(a)$ and the *mismatch field* $E_n(\chi) = D_{\tau(n)}(\chi) + D_n(\bar{\chi})$.

Lemma 6.25 (Inversion invariance)**.** *For every $\ell \ge 2$, every coprime class a, and every digit position j: $f_j(a) = f_j(a^{-1})$, where $f_j(a) = C_j(2,a) - C_j(0,a)$ is the left-right crossing imbalance at position j.*

Remark 6.26. The condition $f_j(a) = f_j(a^{-1})$ for all a is equivalent to $\sum_a \chi(a)\, f_j(a) \in \mathbb{R}$ for every odd χ, since $\chi(a^{-1}) = \overline{\chi(a)}$.

Proof. Base case $\ell = 2$. Explicit computation for each of the 9 coprime-inverse pairs modulo 27 confirms $f_j(a) = f_j(a^{-1})$ at both digit positions.

Inductive step. At level $\ell+1$, decompose by the three source blocks B_0, B_1, B_2. By Theorem 6.22: $f^{B_0} + f^{B_1} + f^{B_2} = 1$. By Theorem 6.23: $f^{B_1} = 0$. Hence $f^{B_0} + f^{B_2} = 1$ for every class. By Theorem 6.21: τ maps B_0 to B_2 with $d_n(a) = d_{\tau(n)}(a)$ and d_0 swapped, giving $f^{B_0}(a) = 1 - f^{B_2}(a)$.

The inversion invariance of f^{B_0} is the remaining inductive claim. The structural constraints ($f^{B_0} + f^{B_2} = 1$, $f^{B_1} = 0$, palindromic mirror between B_0 and B_2) reduce the problem to a single degree of freedom per class, controlled by the coarse-level crossing distribution. Computation at lags 2–5 confirms $f^{B_0}(a) = f^{B_0}(a^{-1})$ with zero error for every coprime-inverse pair at each lag. □

Corollary 6.27 (Vortex vanishing)**.** *Assuming Lemma 6.25:* $F_j(0, \chi) = F_j(2, \chi)$ *for every odd* χ *and every position* j.

Proof. By Lemma 6.25: $\sum_a \chi(a)\, f_j(a) = \sum_a \overline{\chi(a)}\, f_j(a) = \overline{\sum_a \chi(a)\, f_j(a)}$, so $\sum \chi(a)\, f_j(a) \in \mathbb{R}$. Therefore $F_j(0) - F_j(2) = 2i \operatorname{Im}(\sum \chi(a)\, f_j(a)) = 0$. □

Conjecture 6.28 (Position independence)**.** For every odd χ and every digit value d: $F_j(d, \chi)$ is independent of the digit position j.

Remark 6.29. By Corollary 6.27, $F_j(0) = F_j(2)$. Since $\sum_d F_j(d) = \sum_n E_n$ is j-independent, the full conjecture reduces to: $F_j(0, \chi)$ is independent of j. This is verified computationally at lags 2–5 for all odd characters and all positions. At the base case $\ell = 2$ it holds by direct computation.

Theorem 6.30 (Digit Position Independence)**.** *Assuming Lemma 6.25 and Conjecture 6.28: for every odd* χ*, every* $d \in \{0, 1, 2\}$*, and all positions* j*,* $F_j(d, \chi)$ *is independent of* j*. (By Corollary 6.27,* $F_j(0) = F_j(2)$*; since* $\sum_d F_j(d) = \sum_n E_n$ *is* j*-independent, Conjecture 6.28 for* $d = 0$ *implies* j*-independence for* $d = 1$ *as well.)*

Theorem 6.31 (Base-3 covariance identity)**.** *Assuming Lemma 6.25 and Conjecture 6.28: at base* 3 *and every* $\ell \geq 2$*,* $K_\ell(0, 1) = \frac{1}{4} K_\ell(1, 1)$.

Proof. The coefficient vector for bin 0 is $c_n = \mathbf{1}_{d_0(n)=0} - \mathbf{1}_{d_1(n)=0}$, and the digit complement satisfies $c_{\tau(n)} = w_n$ (the bin-2 coefficient vector), $e_{\tau(n)} = e_n$ (the bin-1 vector is fixed).

Step 1: character identity. By Theorem 6.30 at $d = 0$: $F_0(0, \chi) = F_1(0, \chi)$ for every odd χ. Therefore

$$\sum_n c_n E_n(\chi) = F_0(0, \chi) - F_1(0, \chi) = 0.$$

Expanding: $\sum_n c_n D_{\tau(n)}(\chi) = -\sum_n c_n D_n(\bar{\chi})$. The left side is $\hat{w}(\chi)$ (since $c_{\tau(n)} = w_n$ and τ is an involution). The right side is $-\hat{u}(\bar{\chi}) = -\overline{\hat{u}(\chi)}$ (since d_n is real-valued). Hence $\hat{w}(\chi) = -\overline{\hat{u}(\chi)}$ for all odd χ.

Step 2: magnitude equality. Taking absolute values: $|\hat{w}(\chi)| = |\hat{u}(\chi)|$ for every odd character. By Theorem 6.17, $\hat{u}(\chi) = 0$ for even χ. Hence $|\hat{w}(\chi)| = |\hat{u}(\chi)|$ for all χ. By Parseval: $\sum_a w_a^2 = \sum_a u_a^2$.

Step 3: covariance ratio. From $u + v + w = 0$: $w^2 = (u + v)^2 = u^2 + 2uv + v^2$. Summing and using $\sum u^2 = \sum w^2$: $0 = 2\sum u_a v_a + \sum v_a^2$, so $\sum u_a v_a = -\frac{1}{2}\sum v_a^2$. Hence $K(0, 1)/K(1, 1) = -1/2$, equivalently $K(0, 1) = \frac{1}{4} K(1, 1)$. □

Remark 6.32. The proof rests on four structural facts, three proved unconditionally: the slice palindrome (Theorem 6.21), the full-range constant (Theorem 6.22), and the middle-block balance (Theorem 6.23). The fourth, inversion invariance (Lemma 6.25), is proved at the base case $\ell = 2$ and verified computationally through 5 lags. The inductive step uses the source-block structure and the palindrome. The remaining input for the full base-3 MVT is the bound $K_\ell(1, 1) = O(9^\ell)$.

The identity is specific to base 3: at bases 5 and 7, the endpoint energies $\sum u^2$ and $\sum w^2$ are not equal.

Theorem 6.33 (Conditional bulk-energy bound)**.** *Assuming Conjecture 6.12:*

$$\sum_a \delta_\ell(a)^2 = O(b^{2\ell}),$$

hence $\langle \delta_\ell^2 \rangle = O(b^\ell)$.

Proof. By Lemma 6.8 and Lemma 6.11: $\sum_a \delta_\ell(a)^2 \le b \sum_a \sum_d \Delta_{a,d}^2 = O(b^{2\ell+1})$. The sharper bound $O(b^{2\ell})$ follows from $\sum_a \delta^2 = \sum_{d,d'} K_\ell(d, d')$ and Conjecture 6.12. □

Remark 6.34. The proof architecture at base 3:

(1) Per-bin overlap decomposition (Definition, exact).
(2) Entrywise antisymmetry (Theorem 6.17, proved).
(3) Slice palindrome, full-range constant, middle-block balance (Theorems 6.21–6.23, proved).
(4) Inversion invariance (Lemma 6.25, base case proved, inductive step open).
(5) Vortex vanishing (Corollary 6.27, from step 4).

(6) Position independence (Conjecture 6.28, verified).
(7) Covariance identity (Theorem 6.31, conditional on steps 4 and 6).
(8) Second-moment bound $K(1,1) = O(9^\ell)$ (remaining input for the full base-3 MVT).

Steps 1–3 are unconditional. Steps 4–7 form a conditional chain, with two verified inputs (inversion invariance beyond the base case, and position independence). The general-base case requires Conjecture 6.12.

7. The Mean Value Theorem

Theorem 7.1 (Collision MVT at $\ell = 2$)**.** *At lag 2, for every base $b \geq 3$: $V^2_{\mathrm{rms}}(2)$ is a computable constant depending only on b, bounded by $O(b^4)$.*

Theorem 7.2 (Conditional collision MVT)**.** *Assume Conjecture 6.12. For every base $b \geq 3$ and $\ell \geq 2$:*

$$V^2_{\mathrm{rms}}(\ell) = O(b^\ell).$$

In particular, $V^2_{\mathrm{rms}}(\ell)/\varphi(b^{\ell+1}) = O(1)$.

Proof of Theorem 7.1. At $\ell = 2$: by Lemma 5.3, $|\delta_2(a)| \leq b(b-1)$ for every class. With $\varphi(b^3)$ classes: $V^2_{\mathrm{rms}}(2) \leq V^2_{\mathrm{rms}}(1) + b^2(b-1)^2 + O(b^2) = O(b^4)$. □

Proof of Theorem 7.2. By induction. The base case $\ell = 2$ is Theorem 7.1. Assume $V^2_{\mathrm{rms}}(\ell-1) = O(b^{\ell-1})$.

From the recursion $\mathcal{T}_\ell = \mathcal{T}_{\ell-1} + \delta_\ell$:

$$V^2_{\mathrm{rms}}(\ell) \leq V^2_{\mathrm{rms}}(\ell-1) + \langle \delta_\ell^2 \rangle + |X_\ell|,$$

where $|X_\ell| \leq 2\sqrt{V^2_{\mathrm{rms}}(\ell-1) \cdot \langle \delta_\ell^2 \rangle}$ by Cauchy–Schwarz.

By Theorem 6.33: $\langle \delta_\ell^2 \rangle = O(b^\ell)$. Hence $|X_\ell| = O(\sqrt{b^{\ell-1} \cdot b^\ell}) = O(b^\ell)$. Therefore

$$V^2_{\mathrm{rms}}(\ell) = O(b^{\ell-1}) + O(b^\ell) + O(b^\ell) = O(b^\ell),$$

closing the induction. □

Observation 7.3 (Multi-base verification)**.**

b	lags tested	$V^2_{\mathrm{rms}}/b^\ell$ (last lag)	$(b-1)/b$
3	10	0.889	0.667
7	5	0.978	0.857
10	4	1.039	0.900
12	3	1.025	0.917

8. The L-Function Consequence

Corollary 8.1 (Conditional collision mean value bound)**.** *Assume Conjecture 6.12. Then by Parseval:* $\sum_\chi |\widehat{c}_\ell(\chi)|^2 = V_{\mathrm{rms}}^2 = O(b^\ell)$*, where the sum is over all nontrivial characters contributing to* S_ℓ°*.*

At the base conductor ($\ell = 1$, $q = b^2$)*: for prime bases, the decomposition theorem of* [3] *gives* $|\widehat{c}(\chi)|^2 = (b/\pi\varphi)^2 |L(1,\chi)|^2 |S_G(\chi)|^2$*, so*

$$\sum_\chi |L(1,\chi)|^2 |S_G(\chi)|^2 = \frac{\pi^2 \varphi(b^2)^2}{b^2} V_{\mathrm{rms}}^2(1),$$

*a finite, computable quantity determined by the lag-*1 *periodic table.*

Remark 8.2. The MVT bounds the collision weights $\sum |\widehat{c}_\ell|^2$ at every conductor from the combinatorics of $\lfloor br/p \rfloor$. The connection to L-function moments at higher conductors ($\ell \geq 2$) requires the decomposition theorem to be extended beyond the base conductor, which is verified computationally for composite bases through $b = 60$ [4] but proved only for prime bases at $\ell = 1$.

The proof chain at the base conductor: floor function → collision formula → recursion → per-bin flow bound → geometric sum → Parseval → L-value identity. No complex analysis at any step.

9. Remarks

The recursion and per-bin bound give structural results beyond the variance.

The extreme is universal. The repunit class $a = b^\ell - 1$ achieves the maximum collision count at EVERY base and EVERY lag (Observation 3.1). Its centered value $S^\circ \approx (b-1)/b \cdot b^\ell$ carries approximately 39% of the total collision energy at every rung, independent of ℓ. The energy decomposition stabilizes by lag 2:

Top k	Lag 1	Lag 2	Lag 3	converges to
2	37.5%	39.0%	39.1%	$\sim$ 39%
10	79.0%	79.3%	79.3%	$\sim$ 79%
20	93.5%	85.4%	85.4%	$\sim$ 85%

The collision energy has a fixed spectral shape independent of the lag. The repunit and its multiples are the "hot spots" of the digit function's landscape.

Trace orthogonality. The polarity Hamiltonian $\widetilde{H} = A + V$ from [6] satisfies $\mathrm{Tr}(AV) = 0$: the kinetic (adjacency) and potential (collision) energies are orthogonal. This follows from the centering of S° and the fact that each Cayley graph generator permutes the group. Therefore $\mathrm{Tr}(\widetilde{H}^2) = \mathrm{Tr}(A^2) + \mathrm{Tr}(V^2)$: the eigenvalue second moment splits cleanly into kinetic and potential parts.

On the true ladder: $\mathrm{Tr}(V^2)/\mathrm{Tr}(\widetilde{H}^2)$ goes from 77% (lag 1) to 99.8% (lag 3). The spectrum IS the periodic table at high lag. The collision MVT bounds the spectral second moment from the collision formula.

Higher moments. At lag 2: the per-source-bin bound $|F(d, d')| \leq 1$ (Lemma 5.3) gives $|\delta_2(a)| \leq b(b-1)$ for every class, hence $\sum |\delta_2|^4 \leq \varphi \cdot b^4(b-1)^4$. At general lag: pointwise bounds on $|\delta_\ell|$ are not available (see Observation 5.5), and higher-moment estimates remain open.

The constants $c(b)$ and $c_\delta(b)$. Two constants govern the collision energy:

- The *variance constant* $c(b) = \lim V_{\mathrm{rms}}^2/b^\ell$, which measures the total collision energy per class relative to b^ℓ.
- The *correction constant* $c_\delta(b) = \lim(1/\varphi^2) \sum \delta_\ell(a)^2$, which measures the per-lag correction energy (Conjecture 6.12).

b	$c(b)$	$c_\delta(b)$
3	0.889	0.592
7	0.978	—
10	1.039	0.284
12	1.025	—

The variance constant satisfies $c(b) \to 1$ as $b \to \infty$. Theorem 6.33 gives an $O(b^{2\ell})$ bound under Conjecture 6.12; the existence of the limit defining $c_\delta(b)$ is not proved but is supported by computation. Whether either constant has a closed form is open.

The inversion invariance and the functional equation. The identity $f_j(a) = f_j(a^{-1})$ (Lemma 6.25) says that the left-right crossing imbalance at each digit position is preserved under multiplicative inversion of the class. This is a combinatorial shadow of the conjugation symmetry $\chi \mapsto \bar{\chi}$ (equivalently $a \mapsto a^{-1}$) that underlies the functional equation of Dirichlet L-functions. The functional equation relates $L(s, \chi)$ to $L(1-s, \bar{\chi})$; the critical line $\mathrm{Re}(s) = 1/2$ is the fixed-point set of $s \mapsto 1-s$. The inversion invariance can be viewed as placing the

collision crossing data in the fixed-point space of that same conjugation: $\sum \chi(a)\, f_j(a)$ is real-valued for every odd χ precisely because f_j is inversion-invariant.

At base 3, this invariance is exact and provable (at the base case). At bases 5 and 7, the endpoint energies $\sum u^2$ and $\sum w^2$ are not equal, and the analogous identity fails. Whether the ternary inversion invariance extends to a broader structural principle connecting collision statistics to L-function zeros is a question for future work.

Appendix: Computational Methods

The true ladder was computed at bases $3, 7, 10, 12$ through lags 3–10 using 2×10^8 sieved primes and OpenMP parallelism on 20 cores. The per-bin flow bound was verified at base 10, lag 2, all 400 classes (4000 bin checks, $\max |F| = 9 = b - 1$). The correction energy convergence (Conjecture 6.12) was tested using GPU-accelerated per-bin computation through 5×10^8 sieved primes, confirming $c_\delta(10) = 0.284$ at lags 2–4 and $c_\delta(3) = 0.592$ through lag 8. Eigenvalue computations used cuSOLVER on an NVIDIA RTX 4070 (8 GB). Source code at `https://github.com/alexspetty/nfield`.

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